

34.8 A 22nm 16Mb Floating-Point ReRAM Compute-in-Memory Macro with 31.2TFLOPS/W for AI Edge Devices

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Contents

AI-edge devices demand high-precision computation (e.g. FP16 and BF16) for accurate inference in practical applications, while maintaining high energy efficiency (EF) and low standby power to prolong battery life. Thus, advanced non-volatile AI-edge processors [1, 2] require non-volatile compute-in-memory (nvCIM) [3–5] with a large non-volatile on-chip memory, to store all of the neural network’s parameters (weight data) during power-off, and high-precision high-EF multiply-and-accumulate (MAC) operations during compute, to maximize battery life. Among nvCIMs, ReRAM-nvCIM stands out as a promising candidate due to its lowest cost-per-bit (vs. MRAM, PCM, and eFlash), large on-off ratio, and resilience to magnetic-field interference. However, existing nvCIM macros [3–5] do not support floating-point (FP) computation. Implementing a FP-MAC for nvCIM faces challenges, as shown in Fig.34.8.1, in (1) balancing the bit width tradeoff for weight pre-alignment between accuracy and storage, (2) addressing long latency and energy consumption in MAC operations due to the high input bit width in FP format, and (3) managing high array current consumption when accessing numerous memory cells (MCs) for FP operations, particularly in the low-resistance-state (LRS) ReRAM cells.

To address these challenges, we developed: (1) a kernel-wise weight pre-alignment (KWPA) scheme to reduce the accuracy loss due to the data truncation during weight pre-alignment; (2) a rescheduled multi-bit input compression (RS-MIC) scheme to reduce MAC energy and latency with lossless compression; (3) an HRS-favored dual-sign-bit (HF-DSB) weight encoding scheme to reduce ReRAM array current consumption. This work presents a FP 16Mb ReRAM-nvCIM macro utilizing foundry 22nm 1T1R ReRAM devices and achieves 28.7 and 31.2TFLOPS/W using FP16 and BF16 precision. Figure 34.8.2 presents the computing flow for FP ReRAM-nvCIM, featuring a K-WPA scheme. This computing process consists of three key steps: (1) input



pre-alignment; (2) MAC operation between the pre-aligned input mantissa (PA- M_{IN}) and the kernel-wise pre-aligned weight mantissa (PA- M_K); and (3) exponent processing. Due to the high nvCIM capacity, which stores the entire neural network on-chip, the weights can be pre-aligned offline to eliminate the need for on-chip weight alignment. The K-WPA pre-aligns all FP weights offline by assigning the maximum exponent in each kernel as the kernel-shared exponent (E_K). It then aligns each weight's sign and mantissa based on its exponent difference ($E_K - E_i$) to generate a Q-bit PA- M_K . In contrast to the conventional layer-wise weight pre-alignment, the K-WPA preserves more weight data from data truncation during the pre-alignment process due to its fine-grained kernel-wise alignment reference. Before initiating the MAC operation, the E_K are retrieved by reading one row of the ReRAM array and stored in the exponent-processing circuit's register. During the MAC operation, the input pre-alignment circuit fetches A inputs ($A < 1023$, depending on the number of accumulation) in FP16/BF16 format. It assigns the maximum exponent of the input group as the input shared exponent (E_{IN}) and aligns the sign and mantissa of each input based on its exponent difference ($E_{IN} - E_i$) to generate a P-bit PA- M_{IN} . PA- M_{IN} are sent to the ReRAM array for MAC operations with the PA- M_K . The resulting MACVs undergo exponent processing to convert them into a FP format. Finally, they are combined with E_{IN} and E_K to produce the output value in FP32 format.

[Figure 34.8.3](#) illustrates the RS-MIC scheme. The RS-MIC reduces the number of MAC cycles by compressing the multi-bit input without compromising input precision and ADC signal margin. First, the RS-MIC fetches $2b$ inputs from $PA - M_{IN}$ (IN[$a + 1 : a$] group) and reschedules them according to their input values. Inputs with values 11, 10 or 01 are rescheduled and compressed into a 1b input in the IN11, IN10 or IN01 group with $FLAG_{MIC} = 11, 10$ or 01, respectively. This process effectively halves the number of input bits by representing their values within $FLAG_{MIC}$ in their group. Conversely, inputs with a value of 00 are skipped and not sent to the ReRAM array for MAC operation. Within each IN11, IN10 or IN01 group, the inputs are sent to ReRAM array across multiple MAC operation cycles, with a fixed number of WLS activated in each cycle. Please note that the correspondence between input and WL does not change after RS-MIC. The example shows 10002 b inputs (65000 s, 12001 s, 11010 s, and 12011 s) require 64 cycles using a conventional digital bit-serial scheme, while only 12 cycles when using rescheduled inputs. Second, in each MAC operation cycle, the inputs are sent to ReRAM array for MAC operation with PA- M_K to generate a partial MAC value ($pMACV_{MIC}$). The MIC-aware accumulator decompresses the $pMACV_{MIC}$ values and accumulates them across multiple cycles in three stages. In the first stage, the MIC-aware accumulator decompresses $pMACV_{MIC}$ by multiplying it with $FLAG_{MIC}$ to generate a $pMACV$ in cycle- i . $pMACV_{cycle-i} = PMACV_{MIC} \times FLAG_{MIC}$. In the second stage, the MIC-aware accumulator accumulates $pMACV_{cycle-i}$ over multiple cycles within the IN11, IN10 or IN01 group (k cycles) to generate a $pMACV$ of the IN[$2j + 1 : 2j$] Group ($pMACV_{IN[2j+1:2j]} = \sum_{i=0}^k pMACV_{cycle-i}$). In the last stage, the MIC-aware accumulator sums the $pMACV_{IN[2j+1:2j]}$ across input place value to derive the final MACV

$$\left(MACV = \sum_{j=0}^{(P/2)-1} 2^{21} \times pMACV_{[N[2j+1:2j]} \right).$$

[Figure 34.8.4](#) presents the HF-DSB scheme, which reduces the ReRAM array current consumption by decreasing the number of LRS cells accessed without affecting the weight value. In a conventional 2's complement format, only the first bit serves as the sign bit, carrying a negative place value, while the subsequent bits represent magnitude bits with positive place values. The HF-DSB representation ($Y_{(DSB)}$) converts a magnitude bit into a sign bit while maintaining the same weight value as the original coefficient in 2's complement ($X_{(2's)}$). However, the coefficient difference between $Y_{(DSB)}$ and $X_{(DSB)}$, referred to as the conversion term, equals $X[N] \times 2^{N+1}$, where N represents the conversion column that undergoes conversion into a sign column bearing a negative place value. This process is exemplified by two cases for an 8 b PA- M_K . In the first case, with $X_{0(2's)} = -1$, X_0 is "1111111", where $N = 2$ and $X_0[N] = 1$, with the conversion bit underlined. Here, X_0 is transformed into Y_0 by adding 2^3 as the conversion term, resulting in $Y_0 = "00000111"$. This transformation swaps five coefficients from 1 (LRS) to 0 (HRS). In the second case, when $X_{1(2's)} = -103$, X_1 is represented as "10011001", $N = 2$ and $X_1[N] = 0$. The conversion term $0 \cdot 2^3$ yields $Y_1 = 10011001$, leaving the coefficient unchanged. The weights, now in HF-DSB format, along with the HF-DSB flag ($FLAG_{DSB}$) are programmed into the ReRAM array. Prior to the MAC operation, $FLAG_{DSB}$ is retrieved from the ReRAM array and stored in a register. During the MAC operation, inputs from RS-MIC and PA-M in HF-DSB representation are computed in the ReRAM-nvCIM. Due to the reduction in the number of LRS cells accessed, the bit line currents for accumulation are lower compared to those resulting from accessing the weights using a 2's complement format. The 5b-ADC then processes the bit line current into digital $pMACV_{ADC}$. Subsequently, the shifter and sign-aware adder combines $PMACV_{ADC}$ based on the weight's place value and the $FLAG_{DSB}$ of each column to generate the final $pMACV_{MIC}$.

[Figure 34.8.5](#) shows the simulated performance of the proposed schemes using ResNet20 model trained for CIFAR-100 dataset under BF16 precision. The K-WPA suppresses the amount of truncated data during weight pre-alignment process by 1.96 to $2.47\times$ compared to layer-wise weight pre-alignment with a varying number of bits for mantissa after pre-alignment. The RS-MIC reduces the MAC operation cycle count by 4.73 and $1.78\times$ compared to the conventional digital bit-serial multi-bit input and the conventional bit-serial multi-bit input with zero-skip. Lastly, using the HF-DSB weight representation, the average energy consumption of ReRAM array is reduced by $1.31\times$. When considering these schemes together, they collectively improve the EF of ReRAM-nvCIM macro by $1.82\times$.

[Figure 34.8.6](#) presents the measurement results from the 22 nm 16 Mb FP ReRAM-nvCIM macro. The Shmoo plot for the FP ReRAM-nvCIM macro validates a computing latency of 5 ns using $V_{DD} = 0.8$ V. The measured EF reached 31.2TFLOPS/W for a BF16-input, BF16-weight, and FP32-output. In comparison to previous work that employ INT8-input, INT8-weight, and INT24-output [1], this work achieves a 1.86x improvement in FoM, where $FoM = IN\ precision \times W\ precision \times OUT\text{-}ratio \times EF$. Thanks to the high-precision FP MAC operations, the top-1 and top-5 inference accuracy is 69.48 and 91.59 % on a ResNet-20 model with CIFAR-100 datasets. [Figure 34.8.7](#) presents the die photo and summary table.



Figure 34.8.1:
Challenges and proposed FP ReRAM-nvCIM structure.



Figure 34.8.2:
Computing flow of FP ReRAM-nvCIM macro with K-WPA.



Figure 34.8.3:
Illustration of RS-MIC and MIC-aware accumulator.



Figure 34.8.4:
Concept and operation of HF-DSB.



Figure 34.8.5:
Simulated performance of proposed schemes.



Figure 34.8.6:
Measurement result and position chart.



Figure 34.8.7:
Die photo and summary table.

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